Demuestre que:

$$Var\{x\} = \mathcal{E}\{(x-\mu_x^2)\} = \mathcal{E}\{x^2\} - \mathcal{E}^2\{x\}$$

$$Var\{x\} = \mathcal{E}\{x^2 - 2\mu_x x + \mu_x^2\}$$

$$\mathcal{E}\{x^2\} - 2\,\mathcal{E}\{x\mu_x\} + \mathcal{E}\{\mu_x^2\}$$

Nota: $\mathcal{E}\{A\} = \int A\,p(x)\,dx$

$$\mu_x^2 = A\int p(x)\,dx = A$$

Nota: $\mathcal{E}\{x\} = \mu_x$.

$$\mathcal{E}\{\mu_x^2\} = \mu_x^2$$

$$\mathcal{E}\{x^2\} - 2\,\mu_x\mu_x + \mu_x^2$$

$$\mathcal{E}\{\{x^2\} - \{\mu_x^2\}\}$$

$$\mathcal{E}\{x^2\} - \{\mathcal{E}\{x\}\}^2$$

Consideraciones

$$\mathcal{E}\{cte\} = cte$$